

ARM vs x86—The Future of Low-Power Computing

A market analyst doesn't realise that vendors of x86-based chips and ARM-based chips are at each other's throats to get a better share of the market. However, this 'fight' is good for a design engineer or a firm as 'competition brings out the best in products'

DILIN ANAND

Ever since tablets gained market share, design firms have been creating different types of tablets to ensure that they do not lose out on this lucrative market. Soon PC manufacturers too joined the race and dumped their own ARM-based tablets into the market. What these firms failed to realise, or perhaps ignored, was that the growing tablet market was also cannibalising their PC sales.

During the first-quarter earnings call in 2013, Apple CFO Peter Oppenheimer said, "The International Data Corporation (IDC) estimates that the global personal computer market contracted by 6 per cent during the last December quarter." Intel's shares went down from \$29.18 in May 2012 to \$19.51 in December 2012. At the time of writing, these were at \$20.96.

Intel was not going to sit and watch the show. They started pushing out really robust chips that not only perform well but also consume lesser power than the ones before. However, does x86 have what it takes to counter ARM on power consumption front? That would be a game fought on the competitor's turf.

Bring on the competition

Recent advances in ARM architecture with the new 64-bit design, and the growth of applications powered by it, have prompted more vendors to join in and have their own slice. Microsoft has already made its operating system compatible with both x86 and ARM. On the other hand, Advanced Micro Devices (AMD) has tied up with the

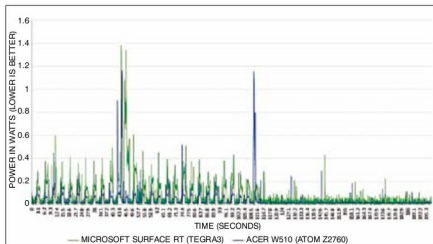


Fig. 1: Power consumption at the CPU and GPU level while the device is idle for a long time (Courtesy: anandtech.com)

UK's ARM and is likely to announce its first ARM-based chip in 2014.

The reason why the 64-bit design is eagerly awaited is that it can help create processors that could handle more random-access memory (RAM) and thus perform memory-intensive tasks, like those in data centres, faster. The weakest link in an RISC-based ARM processor is that it requires far more memory than a CISC-based processor—the 64-bit processor would help to tackle this issue.

So how does this rivalry between x86 and ARM architecture based chip vendors help you as a design engineer or a firm? A quote by David Sarnoff, founder of Radio Corporation of America, gives the answer: "Competition brings out the best in products, and the worst in people."

x86 can get really low on power consumption

Now that Samsung, Apple and

Qualcomm have really powerful ARM-based chips that need very little power to do their work, Intel has started unleashing its own low-power processors. It is not that Intel did not have them before, but the latest ones are really low-power enough to take on ARM chips on their home ground.

Fig. 1 shows power consumption at the CPU and GPU level while the device is lying idle for a long time. The processors in comparison are x86-based Intel Atom Z2760 (based on 32nm lithography) and last-gen ARM-based Tegra 3 (based on 40nm lithography).

"A look at the CPU chart gives us some more granularity, with Tegra 3 ramping up to higher peak power consumption during all of the periods of activity," explains the analyst.

Designers were previously constrained to use ARM-based solutions if they wanted to design a device that